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CLB 321 Laboratory Guide

Batch distillation & two-tank dynamics experiments

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Contents

1. Introduction 1
2. Experimental cycle overview 1
3. Experiment 1: batch distillation 2
4. Experiment 2: Two-tank dynamics 2
5. Preliminary report guidelines 3
6. Presentation guidelines 3
7. Final report requirements 4
8. Timeline diagram 4
9. Appendix ..... a
   - 9.1. Design of Experiments (DOE) Overview ..... a

1. Introduction

This guide frames the CLB 321 laboratory experiments, batch distillation and two-tank dynamic response, as hands-on exercises that link lecture theory to real-world chemical engineering practice. Each experiment follows a five-stage cycle: planning meeting, laboratory session, preliminary report, group presentation and final report. In batch distillation, you will apply mass and energy balances to operate and troubleshoot a simple batch distillation setup.

In the two-tank dynamic response experiment, you will derive and validate first-order plus deadtime (FOPDT) models from time-series level data. By repeatedly testing theoretical models against experimental results and reflecting on discrepancies, you will reinforce key CLB 321 graduate attributes, including problem solving, critical enquiry, teamwork, professional communication and ethical practice, while gaining practical laboratory experience.

2. Experimental cycle overview

For both the batch distillation and two-tank dynamic response experiments, the experimental cycle places ownership of the planning meeting squarely in your hands. Prior to your scheduled laboratory session, your team is advised to organise and document objectives, roles, procedures and materials without direct oversight. Attendance at the laboratory session itself is compulsory for all group members; teaching assistants will record attendance to ensure full participation.

A week after the laboratory session, you will submit your preliminary report via ClickUp; this report must follow the Department of Chemical Engineering's technical report writing guidelines, including clear structure, properly formatted tables and figures, accurate referencing and adherence to prescribed style. At least one week after your submission you will deliver a presentation of your findings in the presence of the course lecturer and a panel of dynamics experts (for example Mr Sonnedecker, Dr Tabane or Ms Motsa) together with teaching assistants; attendance is mandatory and failure to attend will result in forfeiture of the associated marks and/or a direct fail for the two experiments.

Finally, toward the end of the semester, each group will submit a final report for each experiment under the same departmental guidelines; this standardised document will synthesise refined analyses, comprehensive error discussions and concise conclusions.

3. Experiment 1: batch distillation

In this practical, you will work as a single team of eight students divided into two subgroups of four. Subgroup A will perform batch distillation on the isopropyl alcohol and water system using a provided procedure, while Subgroup B will investigate the ethanol-water system, either by replicating the first experiment or by developing an alternative approach subject to teaching assistant approval.

You will then integrate both data sets into one report that adheres strictly to the Department of Chemical Engineering's technical report writing guidelines. As long as you follow those guidelines, you may determine the level of detail and structure in your methods and results sections.

4. Experiment 2: Two-tank dynamics

For the two-tank dynamic response experiment, you will work as a single team of eight students divided into two subgroups of four. Subgroup A will calibrate and characterise the level transmitters and convert their 4 mA to 20 mA signals into actual tank levels, while Subgroup B will impose two-step changes in the manipulated flow rate and record the resulting time series level data.

All experimental details and procedures are provided in the separate instructions document, and any deviations (for example, your choice of step-change magnitudes) require prior approval from the teaching assistant. Attendance at the laboratory session is compulsory for every team member and will be recorded by the teaching assistant. One week after the lab, you must submit your preliminary report via ClickUp in accordance with the Department of Chemical Engineering's technical report writing guidelines.

At least one week later, you will present your findings to the lecturer, teaching assistants and a panel of dynamics experts; failure to attend will forfeit the marks. Finally, toward the end of the semester, you will submit a standardised final report for this experiment, again following departmental style, that integrates your refined FOPDT model fits, detailed

discussions of gain time constant and dead time, and reflections on any discrepancies between theory and practice.

5. Preliminary report guidelines

Your preliminary report, due one week after the laboratory session and to be submitted via ClickUp, should follow the Department of Chemical Engineering's "Guidelines for Writing Technical Reports and Papers". Use a clear, legible font (for example Arial 12, Times New Roman 14 or similar), with 1½-line spacing and double spacing between paragraphs. Organise your report according to the prescribed structure (Table 1 on the guidelines): cover page, title page, synopsis and keywords, introduction, literature review (theory), experimental, results or results and discussion, discussion, conclusions or conclusions and recommendations, recommendations, references and appendices. Acknowledgements, nomenclature lists and appendices are optional.

In the experimental section (Chapter 3.6.1 in the guidelines) describe apparatus, planning and methods with enough detail to allow replication, without using an idiot recipe style. Number and caption all tables, figures and equations according to the style guide, and ensure all units, symbols and references conform to departmental conventions. Failure to meet these formatting and structural expectations may result in deductions under the department's editorial-care criteria.

6. Presentation guidelines

For each experiment, there will be a dedicated presentation session. The full eight-member team must attend both sessions; any absence will forfeit the marks for that session. In the batch distillation session, you will have up to twenty minutes to present your results, immediately followed by an open-ended question period. In the two-tank dynamic response session, you will likewise have up to twenty minutes for your talk and an unrestricted question period thereafter.

Prepare clear, uncluttered slides with text no smaller than 24 pt and well-labelled figures or tables. Begin each presentation with a succinct statement of objectives, highlight only your key findings and conclude with concise take-home messages. Rehearse together to ensure smooth transitions between speakers and confident responses to questions.

7. Final report requirements

The final report, due at the end of the semester via ClickUp, should build on your preliminary report submission (see “Preliminary Report Guidelines”). You must address all comments from both the preliminary report review and your presentation, correcting any issues in clarity, formatting or analysis. Aside from those revisions, the report follows the same departmental structure and style, including cover page, introduction, experimental section, results and discussion, conclusions and references, and complies with the Guidelines for Writing Technical Reports and Papers.

Tables, figures and equations should be numbered and captioned according to the style guide, and the experimental section again must avoid recipe-style descriptions. Failure to incorporate feedback or maintain the required format may result in deductions under the department’s editorial care criteria.

8. Timeline diagram

Figure 1 provides a concise snapshot of the workflow for both batch distillation and two-tank dynamic response experiments. It shows how each stage, starting with the planning meeting, then the laboratory session, the preliminary report, the group presentation and the final report, fits together in a unified view. This visual summary makes it easy to see the overall sequence at a glance.

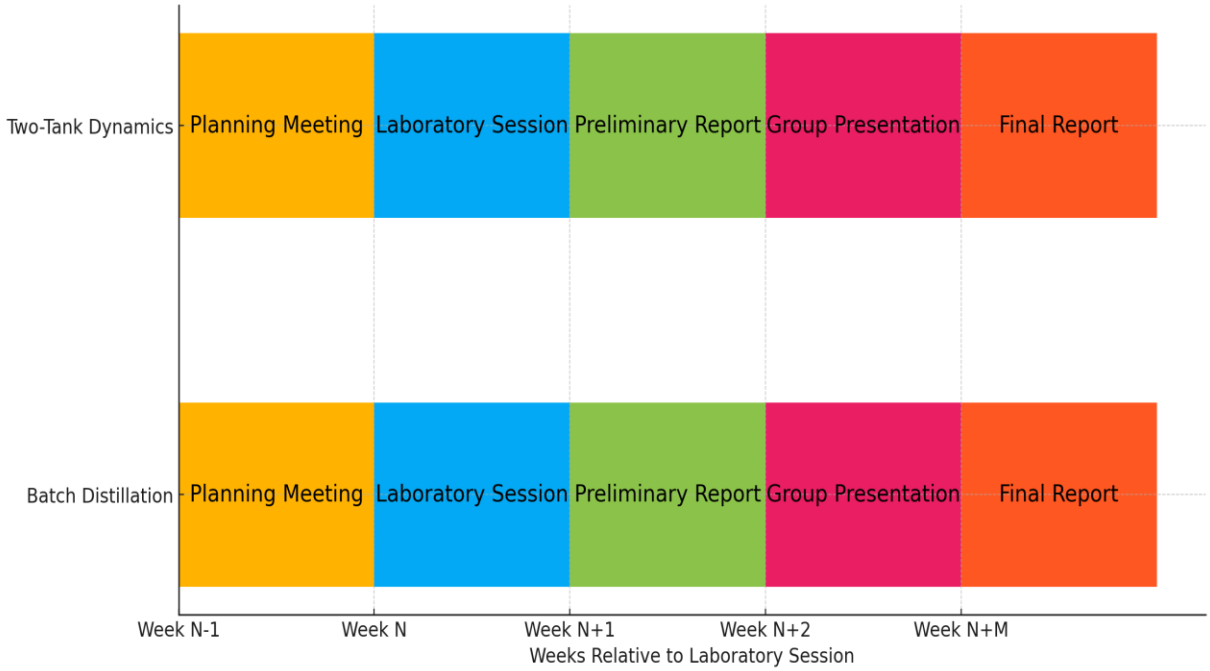


Figure 1: Workflow snapshot for batch distillation and two-tank dynamic response experiments.

9. Appendix

9.1. Design of Experiments (DOE) Overview

Design of experiments (DOE) is a systematic approach to planning experiments so that data can be analysed to yield valid and objective conclusions. DOE emphasises varying multiple factors simultaneously in a structured way, rather than one-at-a-time, to efficiently explore the effect of factors and their interactions on a response. Classic texts such as Montgomery's Design and Analysis of Experiments and the NIST/SEMATECH e-Handbook of Statistical Methods cover these principles in detail (Montgomery, 2017).

9.1.1 One-factor-at-a-time (OFAT)

One-Factor-at-a-Time (OFAT) is the simplest experimental strategy: only one factor is varied at a time while keeping all others constant. The appeal of OFAT lies in its simplicity and intuition; it seems straightforward to isolate the effect of each factor by changing it one by one. However, this method is notoriously inefficient and can be misleading. OFAT requires many runs to explore multiple factors, and more importantly, it fails to detect interactions between factors(*Lesson 1: Introduction to Design of Experiments*).

Ronald Fisher famously argued that studying factors together in a factorial design yield more information than OFAT with the same number of trials. In practice, OFAT might be used for preliminary troubleshooting or demonstration experiments, but DOE practitioners usually prefer factorial designs to glean interactions and optimise efficiency.

9.1.2 Screening Designs (Plackett–Burman, Taguchi)

When many candidate variables exist and time is short, screening designs identify the few that matter. Plackett–Burman matrices use two levels and a small number of runs to estimate main effects while assuming interactions are negligible (Plackett and Burman, 1946) . Taguchi orthogonal arrays serve a similar purpose in quality engineering, balancing economy of runs with the need to separate factor effects; they are often coupled with noise-factor arrays for robustness (Taguchi, 1987; Antony, 2023)

9.1.3 Full factorial designs.

A full factorial tests every combination of factor levels, giving clean estimates of all main effects and interactions. This completeness is valuable when only a handful of factors are

under study or when interaction effects are central to the question. The drawback is that the run count grows exponentially with each added factor (Montgomery, 2017; Box, Hunter and Hunter, 2005).

9.1.4 Fractional factorial designs.

To cut costs, a fractional factorial selects a structured subset of runs from the full factorial. Higher-order interactions are deliberately confounded so that main effects and low-order interactions can still be estimated with far fewer experiments. The key is choosing an appropriate resolution so that the effects you care about are not aliased(Plackett and Burman, 1946)

9.1.5 Response surface designs (Central Composite, Box–Behnken).

Once important factors are known and curvature is suspected, response surface methodology fits second-order models to locate optima. Central Composite Designs augment a factorial core with centre and axial points to estimate quadratic terms (Box & Wilson, 1951). Box–Behnken designs place points at mid-edges and centres, avoiding extreme corners and often requiring fewer runs than a comparable CCD (Myers, Montgomery and Anderson-Cook, 2016; Box, Hunter and Hunter, 2005).

9.1.6 Blocking, randomisation and replication.

Randomisation protects against hidden time trends or drift by scrambling run order (Fisher's principle). Replication provides an estimate of experimental error and tests repeatability. Blocking groups runs under similar nuisance conditions (for example different lab days) so that this known variability can be removed from the error term and precision improved. These three practices underpin all DOE types and safeguard the validity of conclusions (Box, Hunter and Hunter, 2005; Montgomery, 2017).

9.1.7 References

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